TECHNICAL DOCUMENTATION

AI FITNESS COACH ● ENGINEERING

AI Coaching Layer Architecture

A Retrieval-Augmented Generation system for adaptive, evidence-based virtual fitness coaching — covering knowledge retrieval, personalized program generation, session memory, and real-time coaching.

PROJECT	MODULE	DOCUMENT VERSION	LAST UPDATED
AI Fitness Coach	backend/app/ai/	1.0	July 2026

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1 Executive Summary

The AI Coaching Layer is the intelligence core of the AI Fitness Coach application. Rather than generating static workout plans, the system operates as an adaptive virtual coach: it interprets user profiles, produces personalized training programs, delivers real-time guidance during exercise sessions, records workout history, and continuously refines future recommendations based on observed outcomes.

The module is built on a **Retrieval-Augmented Generation (RAG)** architecture that combines Large Language Models (LLMs), semantic search, and structured fitness knowledge to produce reliable, context-aware responses while minimizing hallucination. It is implemented as a set of independent Python services that integrate with the backend through FastAPI, remaining fully decoupled from the presentation and voice-processing layers.

4	5	3	8
CORE AI CAPABILITIES	PUBLIC SERVICES	PRIMARY DATA ENTITIES	WORKFLOW STAGES

DESIGN PRINCIPLE

Fitness knowledge is retrieved from a curated, validated knowledge base rather than generated solely from the language model's internal parameters. This grounds every recommendation in evidence-based content and materially reduces the risk of unsafe or fabricated guidance.

2 System Overview

The AI system is composed of four major capabilities that together transform the application from a static workout planner into an intelligent coaching assistant capable of learning from every completed training session.

Knowledge Retrieval

Semantic search over a curated fitness knowledge base (RAG).

Program Generation

Personalized workout plans built from profile, goals, and history.

Session Memory

Adaptive memory of completed sessions to inform future programs.

Live Coaching

Real-time guidance, feedback, and safety monitoring during workouts.

Each capability is implemented as an isolated service, allowing the system to evolve independently across knowledge ingestion, generation logic, and real-time interaction without introducing cross-module coupling.

3 Technology Stack

The AI Coaching Layer is built on a focused, production-oriented technology stack, selected for low-latency inference, vector-native storage, and seamless integration with the existing backend and mobile client.

TABLE 3.1 — CORE TECHNOLOGY STACK

Component	Technology
Backend	`FastAPI (Python 3.12)`
Database	`PostgreSQL + pgvector`
Large Language Model	`Groq API (Llama-3.3-70B)`
Embeddings	`FastEmbed (BAAI/bge-small-en-v1.5)`
Mobile Client	`Flutter`
Deployment	`Docker Compose`

NOTE

pgvector extends PostgreSQL with native vector similarity search, allowing embedded knowledge, exercise data, and session history to be queried within the same relational database rather than a separate vector store.

4 Overall Architecture

The system follows a layered architecture in which the Flutter mobile client communicates exclusively with the FastAPI backend, which in turn delegates all intelligence-related operations to the AI Coaching Layer. The AI layer communicates with the backend through direct Python service calls, while the backend exposes REST APIs consumed by the Flutter application — keeping the client fully decoupled from AI implementation details.

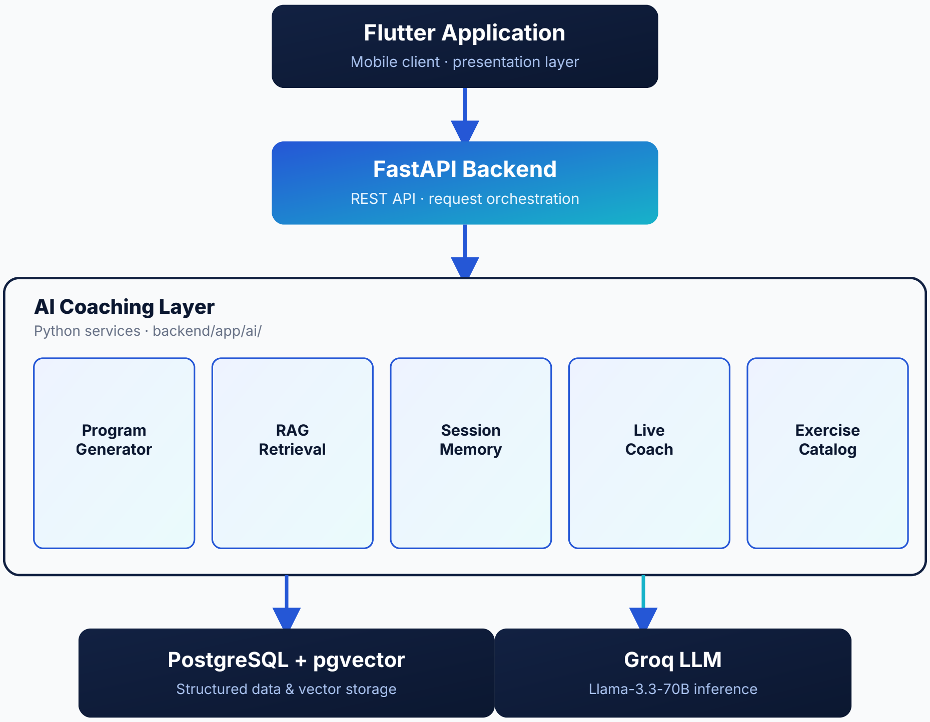

Figure 4.1 — End-to-end request flow from the Flutter client through the AI Coaching Layer to persistence and inference services.

This structure isolates concerns cleanly: the Flutter application handles presentation only; the FastAPI backend handles request orchestration and persistence access; and the AI Coaching Layer encapsulates all reasoning, retrieval, and generation logic behind five internal services, backed by PostgreSQL with pgvector for storage and the Groq-hosted Llama-3.3-70B model for inference.

5 AI Components

The AI Coaching Layer is composed of five internal services, each responsible for a distinct capability within the overall coaching experience. This section describes each component's purpose, inputs, and behavior.

5.1 Retrieval-Augmented Generation (RAG)

The RAG module supplies factual fitness knowledge to the language model prior to response generation. Rather than relying solely on the LLM's internal knowledge, relevant information is retrieved from a curated knowledge base using semantic similarity search. The retrieved context is injected into the prompt, producing more accurate and trustworthy recommendations.

- `Document embedding`
- `Semantic search`
- `Knowledge retrieval`
- `Context injection`
- `Hallucination reduction`

5.2 Exercise Catalog

The Exercise Catalog stores structured information describing every supported exercise. Each entry includes:

- Name
- Equipment
- Difficulty
- Muscle groups
- Instructions
- Safety notes
- Progressions
- Regressions

During workout generation, the AI selects exercises directly from this catalog rather than inventing new ones, ensuring both consistency and safety across generated programs.

5.3 Personalized Program Generation

The program generation service constructs complete workout plans using the following inputs:

- User profile
- Fitness goals

- Available equipment
- Physical limitations
- Retrieved fitness knowledge
- Previous workout history (optional)

The generated program conforms to the existing backend schema, allowing it to be persisted directly without additional transformation.

5.4 Personal Session Memory

After each completed workout, a summary of the session is generated and embedded into the vector database. The stored memory includes:

- Completed exercises
- User feedback
- Fatigue level
- Session difficulty
- Notes
- Performance indicators

This memory allows the AI to adapt future programs according to the user's ongoing progression.

5.5 Live AI Coach

The Live Coach provides assistance throughout the workout session, structured around four stages:

1. Check-In	2. Exercise Feedback	3. Mid-Session Coaching	4. Session Summary
Session start and readiness assessment.	Guidance during individual exercises.	Ongoing monitoring and adjustment.	Wrap-up and performance recap.

During a live session, the AI can:

- Answer user questions
- Monitor fatigue
- Recommend additional rest
- Suggest exercise regressions
- Detect potential safety concerns
- Encourage the user throughout the session

6 Knowledge Base

The knowledge base is built from trusted fitness resources. Sources include:

- Curated coaching documents
- Scientific guidelines
- Exercise metadata
- Selected research papers

QUALITY ASSURANCE

Only validated documents are indexed into the knowledge base, ensuring that AI-generated responses remain evidence-based and consistent across sessions.

7 Data Model

The AI layer introduces three primary data entities that support retrieval, generation, and personalization.

TABLE 7.1 — PRIMARY DATA ENTITIES

Entity	Description
Knowledge Chunks	Stores embedded fitness knowledge for semantic retrieval.
Exercises	Stores the complete structured exercise catalog.
User History	Stores embedded summaries of completed workout sessions to enable personalized recommendations.

8 AI Workflow

The AI Coaching Layer operates as a continuous feedback loop. Each completed session informs the generation of future programs, allowing the coach to improve its recommendations over time.

Figure 8.1 — The eight-stage AI workflow, operating as a continuous feedback loop from user profile to adapted future programs.

1. User profile is loaded.
2. Relevant knowledge is retrieved.
3. Exercise catalog is queried.
4. The LLM generates a personalized workout.
5. The workout is performed.
6. User feedback is collected.
7. Session history is indexed.
8. Future programs are adapted using previous performance.

9 Public Services

The AI layer exposes a set of reusable services consumed by the backend API. These services remain fully independent of the frontend implementation.

TABLE 9.1 — EXPOSED AI SERVICES

Service	Purpose
Generate Program	Creates personalized workout plans.
Suggest Next Program	Generates the next adaptive session.
Retrieve Knowledge	Performs semantic search over the knowledge base.
Index Session	Stores completed workout history.
Live Coach	Provides real-time coaching during a session.

10 Configuration

The module requires the following environment configuration variables:

- `GROQ_API_KEY`
- `DATABASE_URL`
- `CHAT_MODEL`
- `EMBEDDING_MODEL`
- `EMBEDDING_DIMENSIONS`

DOCKER COMMANDS

```
# Build and start the backend, database, and AI services
docker compose up --build

# Run in detached mode
docker compose up -d

# Stop all services
docker compose down
```

Docker Compose is used to manage the backend services and database environment as a single, reproducible deployment unit.

11 Testing & Validation

The implementation includes dedicated smoke tests that validate the complete AI pipeline prior to deployment. Coverage includes:

- Knowledge ingestion
- Retrieval quality
- Workout generation
- History indexing
- Personalized recommendations
- Live coaching

VALIDATION SCOPE

These tests exercise each stage of the AI workflow described in Section 8, confirming that retrieval, generation, and adaptation behave correctly end-to-end before a release is promoted.

12 Current Implementation Status

Completed

- RAG knowledge retrieval
- Exercise catalog
- Workout generation
- User memory system
- Adaptive recommendations
- Live AI coaching
- Scientific knowledge ingestion
- Docker integration

Remaining

- FastAPI endpoint integration
- Flutter interface implementation
- Voice interaction (Speech-to-Text / Text-to-Speech)

13 Future Improvements

Beyond completing the remaining integration work outlined in Section 12, the following areas represent the natural next steps for extending the AI Coaching Layer:

- **FastAPI endpoint integration** — exposing the AI services described in Section 9 through complete, production-ready REST endpoints.
- **Flutter interface implementation** — connecting the mobile client to the AI Coaching Layer's generation and live coaching services.
- **Voice interaction** — adding Speech-to-Text and Text-to-Speech capabilities to the Live AI Coach for hands-free use during workouts.
- **Wearable device integration** — incorporating physiological data to further personalize program generation and live coaching.
- **Advanced performance analytics** — expanding session memory into longer-term progression insights for the user.

14 Conclusion

The AI Coaching Layer transforms the application from a conventional fitness tracker into an intelligent virtual coach. By combining Retrieval-Augmented Generation, semantic search, structured exercise data, and personalized session memory, the system delivers adaptive, evidence-based guidance throughout the user's fitness journey.

Its modular architecture ensures scalability, maintainability, and seamless integration with the backend API and mobile application, while providing a strong foundation for future features such as voice interaction, wearable device integration, and advanced performance analytics.

End of document — AI Fitness Coach, AI Coaching Layer Technical Documentation, Version 1.0, July 2026.